

Mini Project Start

Project Name:

Product Review Sentiment Analyzer – E-commerce

Dataset:

Amazon Product Reviews Dataset (10,000 reviews)

Features (X Columns):

1. Review Text
2. Review Length (number of words/characters)
3. Verified Purchase Status
4. Product Rating (1–5 stars)

Target (y Column):

Sentiment Category

- Positive
- Neutral
- Negative

Question the Model Will Answer:

Given a customer's product review, can the model automatically determine whether the sentiment expressed is Positive, Neutral, or Negative?

Project Overview

M1: Data Preparation & Exploration

- Load 10,000 Amazon reviews
- Perform text cleaning:
 - Lowercasing
 - Remove punctuation
 - Remove special characters
 - Remove stopwords using NLTK
- Explore review rating distribution
- Visualize sentiment and rating trends

M2: Machine Learning Model

- Convert review text into numerical features using **TF-IDF Vectorizer**
- Train a **Naive Bayes Classifier**
- Predict:
 - Positive
 - Neutral
 - Negative
- Save trained model as:
 - sentiment_model.pkl
 - tfidf_vectorizer.pkl

M3: Web Application

- Build a Flask application
- User pastes a review into a form
- Display:
 - Predicted sentiment
 - Confidence percentage
 - Highlighted words contributing to prediction

M4: Deployment

- Deploy application online
- Generate live URL
- Demonstrate sentiment prediction with sample reviews

Technology Stack:

- **Python**
- **NLTK**
- **TF-IDF Vectorizer**
- **Naive Bayes**
- **Flask**
- **HTML**
- **CSS**
- **JavaScript**

- **SQLite**